



वित्तीय सेवाएं विभाग
DEPARTMENT OF
FINANCIAL SERVICES

PSBs Hackathon Series 2026

(An Initiative of Government of India, Ministry of Finance , Department of Financial Services)



PRESENTS



A i - C S P A R C

AI Powered Innovation for Customer: Security, Privacy, Assurance, Reliance, Convenience

Digital Trust | Ease of doing Business

Promoting Innovation and Fostering Collaboration



SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaia School of Engineering
(formerly K J Somaia College of Engineering)

PROBLEM STATEMENT & TEAM DETAILS

TEAM NAME (AS SUBMITTED ON PORTAL)		GenZPT		
PROBLEM STATEMENT TITLE		AI Driven Early Warning System for Internal & Privileged User Fraud		
	NAME	GENDER	AREA OF EXPERTISE	
Member 1	Abhaykumar Roy	Male	Backend Developer	
Member 2	Dhiraj Agrawal	Male	AiML	
Member 3	Aditya Thakare	Male	Frontend Developer	
Member 4	Sanyam Patwari	Male	Banking & Security	

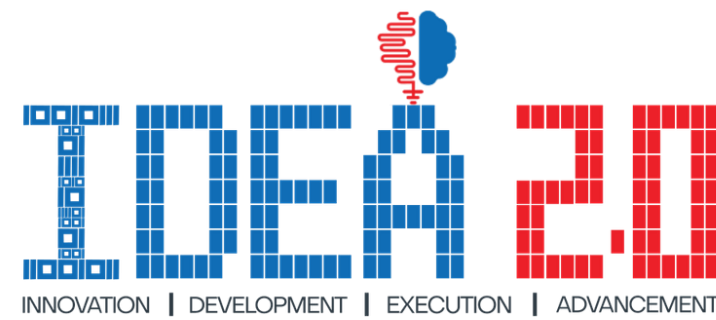


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यूनियन बैंक
ऑफ़ इंडिया
भारत सरकार का उपक्रम

Union Bank
of India
A Government of India Undertaking



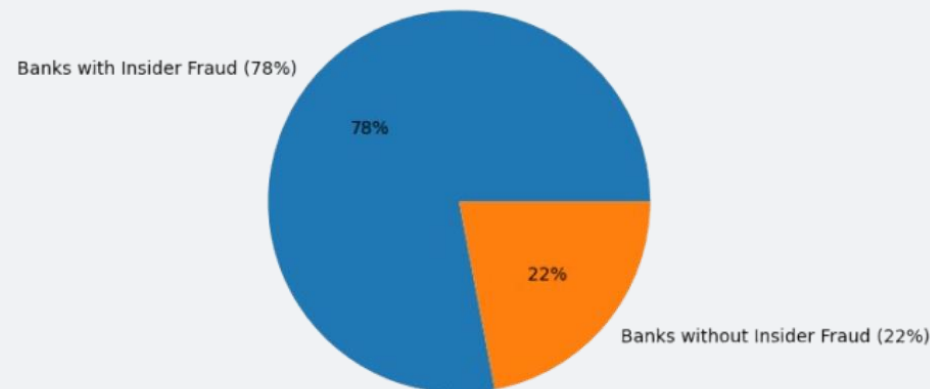
SurakshaAI



THE CHALLENGE

Current Problem Landscape:

- **Problem:** 78% of banks faced insider fraud in last 12 months. Detection takes 287 days.
- Cost: \$15.38M per incident.



THE SOLUTION

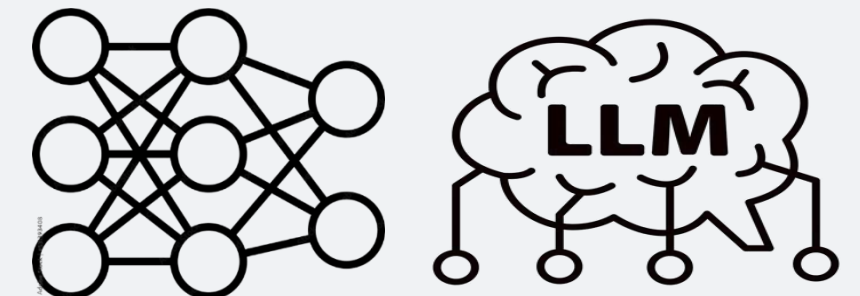
- AI-powered system that monitors privileged users across banking systems and detects fraud in real-time.
- Real-time monitoring of core banking, treasury, loan systems
- Behavioral baselining per user
- Anomaly detection with risk scores
- Alerts with contextual explanations
- Dashboard for fraud investigation team



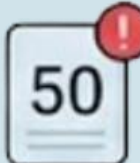
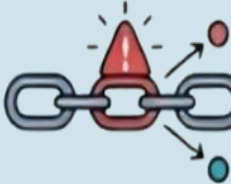
INNOVATION

Uniqueness:

First solution combining **Graph Neural Networks** (detects multi-user collusion) + **LLM Assistant** (natural language investigation queries)



OUTLINE OF UNIQUE & INNOVATIVE SOLUTION

Existing Solutions	Our Solution
 Individual user monitoring	 Graph Neural Network GNN Collusion Detection – catches fraud rings
 Manual log analysis	 Large Language Model LLM Assistant – natural language queries
 Static risk scores	 Risk Propagation – connected users auto-updated
 Generic approach	 Privileged User Focus – specialized models

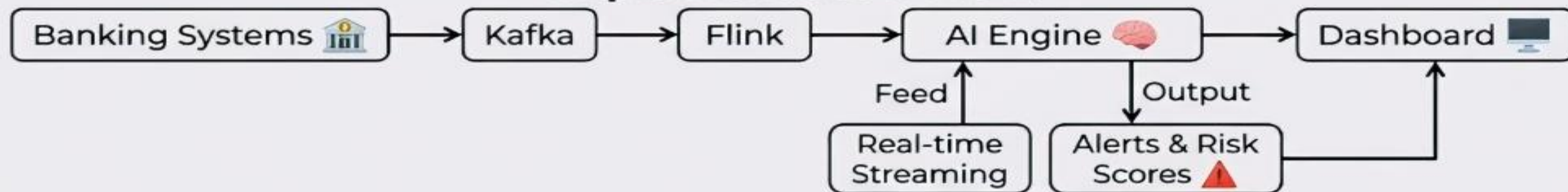


Innovation: No commercial platform offers **GNN-based collusion detection + LLM investigation** in one system.

TECHNICAL APPROACH

Layer	Technology
Data Ingestion	Kafka
Stream Processing	Apache Flink
ML Models	Python, PyTorch, PyTorch Geometric, scikit-learn
LLM	Gemini 2.5 Flash API
Storage	Elasticsearch, PostgreSQL
Dashboard	React, WebSocket

Implementation Flow:



- **Data Source:** CMU CERT Dataset (135M events, 4,000 users, validated threat scenarios)

FEASIBILITY & VIABILITY

Aspect	Assessment
Data	CMU CERT dataset provides realistic banking logs
Technology	All tools are production-proven (Kafka, PyTorch, Elasticsearch)
Models	Autoencoders reduce false positives by 59% (proven in academic papers)

Key Challenges & Mitigations:

Challenge	Mitigation
Real-world log formats	Adapter pattern for easy integration
GNN complexity	Start with simpler graph algorithms, then scale
False positives	Human-in-the-loop feedback + continuous retraining

IMPACT & BENEFITS

Audience

- **CISO:** Real-time visibility into threats
- **Investigator:** Investigation time reduced from 4hrs to 30min
- **Compliance Officer:** Provides audit-ready trails for regulations

Economic

- **Fraud Loss:** 80% reduction, from \$15.38M to \$3M
- **Investigation Time:** Decreased by 87%
- **ROI:** Achieved 7.6x within the first year

Social

- **Customer Trust:** Increased by 40%
- **Employee Privacy:** Ensured through role-based monitoring
- **Investigator Well-being:** Improved with 60% fewer false alerts

Environment

- **Energy Efficiency:** Serverless architecture scales on demand, reducing consumption
- **Paperless:** 100% digital case management eliminates paper waste
- **Carbon Footprint:** Reduced through optimized processes and remote team support

“80% fraud reduction, 87% faster investigations, 7.6x ROI—while protecting customer trust, privacy, and the environment.”

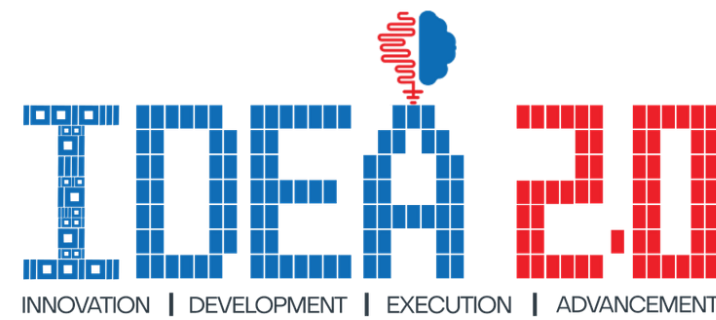


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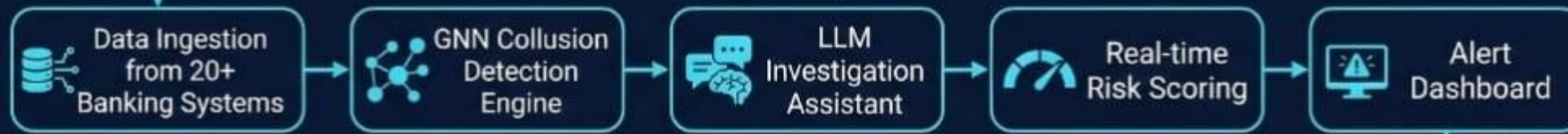
BUSINESS MODEL

AI Insider Threat Detection System - Business Model

Inputs/Resources



Platform Components



Revenue Flow



Value Delivered



Value Proposition

- GNN collusion detection
- LLM investigation assistant
- Real-time monitoring 60% reduction in false positives vs. rule-based systems
- Explainable AI

Target Customers

- Tier 1 Banks
- Regional Banks
- Fintech Companies
- Decision Makers(CISO)

RESEARCH & REFERENCES

1] AI-Driven Insider Risk Management with Adaptive Scoring (2025)

This paper proposes an AI-based system that detects insider threats using **behavioral analysis + dynamic risk scoring**. It improves detection accuracy and reduces false positives using deep learning models and continuous feedback learning. (<https://arxiv.org/abs/2505.03796>)

2] Facade: Deep Contextual Anomaly Detection for Insider Threats (2024)

This system (used at Google) detects insider threats by analyzing **user actions along with context (who, what, when, why)**. It achieves extremely low false positives using deep learning and contextual modeling. (<https://arxiv.org/abs/2505.03796>)

3] Insider Threat Detection using GCN + Bi-LSTM (2025)

This paper combines **Graph Neural Networks (GNN)** with **temporal models (LSTM)** to detect insider threats. It models both relationships between users and time-based behavior patterns for highly accurate detection. (<https://ieeexplore.ieee.org/abstract/document/11311570/>)

4] Explainable AI in Insider Financial Fraud Detection (2026)

This paper explains how **Explainable AI (SHAP, LIME)** improves fraud detection systems by making them transparent and trustworthy. It also compares deep learning models like LSTM, CNN, and GNN. (<https://dl.acm.org/doi/full/10.1145/3794859>)

- Summary for the above Problem Statement :-

<https://drive.google.com/file/d/13rmvefZ6oKO9IqYP-g06BAGWaLo4k-2q/view?usp=sharing>